Class 9 Mathematics — Exercise 4.3

Basic Concept

(1) HCF (Highest Common Factor)

The HCF of two or more expressions is the **product of all common factors with least power**.

(2) HCF of Monomials

- Find the HCF of the **numerical coefficients** by prime factorization.
- For each variable, take the **least power** that appears in all monomials.
- HCF = product of these.

Example: $16 = 2^4$, $9 = 3^2$ $\Rightarrow$ no common prime factor, so HCF of 16 and 9 is 1.

(3) HCF by Factorization

Factorize each expression completely, then pick out the **common factors** (with least power). Their product is the HCF.

Question 1(a)

Find the HCF of the following monomials by completing the table.

Monomials	HCF of coeff.	HCF of p	HCF of q	HCF of r	HCF
$16p^3q,\ 9pq^2r$	1	p	q	—	pq
$10p^3q^2r,\ 5p^2qr,\ 15p^2qr^2$	5	p^2	q	r	$5p^2qr$
$14p^4qr^4,\ 28p^3qr^2,\ 7p^2qr^2,\ 21p^2q^2r^4$	7	p^2	q	r^2	$7p^2qr^2$

Working for coefficients:

- $16 = 2^4$, $9 = 3^2$ $\Rightarrow$ HCF $= 1$
- $10 = 2 \times 5$, $5 = 5$, $15 = 3 \times 5$ $\Rightarrow$ HCF $= 5$
- $14 = 2 \times 7$, $28 = 2^2 \times 7$, $7 = 7$, $21 = 3 \times 7$ $\Rightarrow$ HCF $= 7$

Question 1(b)

If all common factors with least power of three unknown polynomials are 2^2, 3, pq and $(p+q)^2$, then what would be their HCF?

HCF = product of all common factors with least power:

$$\begin{aligned} \text{H.C.F} &= (2^2)(3)(pq)(p+q)^2 \\ &= (4)(3)(pq)(p+q)^2 \end{aligned}$$

$$= \boxed{12pq(p+q)^2} \text{ Ans}$$

Question 1(c)

Write any two polynomials of your choice having HCF as 1.

$$\boxed{(x+y)^2, \quad (x-y)^2} \text{ Ans}$$

because $(x+y)^2 = (x+y)(x+y)$ and $(x-y)^2 = (x-y)(x-y)$ have **no common factor**, so their HCF is 1.

Question 1(d)

The only common factor of two polynomials is $m-n$ and the only uncommon factor is m^2+n^2. Can you guess the unknown polynomials?

Common $= m-n$, Uncommon $= m^2+n^2$

$$\boxed{(m-n), \qquad (m-n)(m^2+n^2)} \text{ Ans}$$

Question 1(e)

Can you guess the HCF of two polynomials x^3+5x+1 and $1+5x+x^3$ without any procedure?

Both polynomials have exactly the same terms (x^3, $5x$ and 1), only written in a different order:

$$1+5x+x^3 = x^3+5x+1$$

So both polynomials are the same, and the HCF is the polynomial itself:

$$\boxed{\text{H.C.F} = x^3+5x+1} \text{ Ans}$$

Question 2

Find the HCF by factorization: $(x+y)^2, \quad x^2-y^2$

$(x+y)^2$	x^2-y^2
$= \boxed{(x+y)}(x+y)$	$= \boxed{(x+y)}(x-y)$

$$\text{H.C.F} = \boxed{x+y} \text{ Ans}$$

Question 3

$$(a-b)^3, \quad a^2-2ab+b^2$$

$(a-b)^3$	$a^2-2ab+b^2$
$=\boxed{(a-b)(a-b)}(a-b)$	$=(a-b)^2$
	$=\boxed{(a-b)(a-b)}$

$$\text{H.C.F} = (a-b)(a-b)$$
$$= \boxed{(a-b)^2 = a^2-2ab+b^2} \text{ Ans}$$

Question 4

$$a^3b-ab^3, \quad a^5b^2-a^2b^5$$

a^3b-ab^3	$a^5b^2-a^2b^5$
$=ab(a^2-b^2)$	$=a^2b^2(a^3-b^3)$
$=\boxed{ab}(a+b)\boxed{(a-b)}$	$=\boxed{(ab)}(ab)\boxed{(a-b)}(a^2+ab+b^2)$

$$\text{H.C.F} = \boxed{ab(a-b)} \text{ Ans}$$

Question 5

$$x^2-49, \quad x^2-4x-21$$

For $x^2-4x-21$: $-21 \to -7$ and $3 \quad (-7+3=-4)$

x^2-49	$x^2-4x-21$
$=x^2-7^2$	$=x^2-7x+3x-21$
$=(x+7)\boxed{(x-7)}$	$=x(x-7)+3(x-7)$
	$=\boxed{(x-7)}(x+3)$

$$\text{H.C.F} = \boxed{x-7} \text{ Ans}$$

Question 6

$$12x^2+x-1, \quad 15x^2+8x+1$$

For $12x^2+x-1$: $-12 \to 4$ and $-3 \quad (4-3=1)$
For $15x^2+8x+1$: $15 \to 5$ and $3 \quad (5+3=8)$

$12x^2 + x - 1$	$15x^2 + 8x + 1$
$= 12x^2 + 4x - 3x - 1$	$= 15x^2 + 5x + 3x + 1$
$= 4x(3x + 1) - 1(3x + 1)$	$= 5x(3x + 1) + 1(3x + 1)$
$= \boxed{(3x + 1)}(4x - 1)$	$= \boxed{(3x + 1)}(5x + 1)$

$$\text{H.C.F} = \boxed{3x + 1} \text{ Ans}$$

Question 7

$$c^2x^2 - d^2, \quad acx^2 - bcx + adx - bd$$

$c^2x^2 - d^2$	$acx^2 - bcx + adx - bd$
$= (cx)^2 - (d)^2$	$= cx(ax - b) + d(ax - b)$
$= \boxed{(cx + d)}(cx - d)$	$= (ax - b)\boxed{(cx + d)}$

$$\text{H.C.F} = \boxed{cx + d} \text{ Ans}$$

Question 8

$$m^2 - n^2, \quad m^4 - n^4, \quad m^6 - n^6$$

First:

$$m^2 - n^2 = (m + n)(m - n)$$

Second:

$$\begin{aligned} m^4 - n^4 &= (m^2)^2 - (n^2)^2 \\ &= (m^2 + n^2)(m^2 - n^2) \\ &= (m^2 + n^2)(m + n)(m - n) \end{aligned}$$

Third:

$$\begin{aligned} m^6 - n^6 &= (m^3)^2 - (n^3)^2 \\ &= (m^3 + n^3)(m^3 - n^3) \\ &= (m + n)(m^2 - mn + n^2)(m - n)(m^2 + mn + n^2) \\ &= (m + n)(m - n)(m^2 - mn + n^2)(m^2 + mn + n^2) \end{aligned}$$

Common factors: $(m + n)$ and $(m - n)$

$$\text{H.C.F} = (m + n)(m - n) = \boxed{m^2 - n^2} \text{ Ans}$$

Question 9

$$ax^2 + 2a^2x + a^3, \quad 2ax^2 - 4a^2x - 6a^3, \quad 3(ax + a^2)^2$$

First:

$$ax^2 + 2a^2x + a^3 = a(x^2 + 2ax + a^2)$$
$$= a(x + a)^2 = \boxed{a(x + a)}(x + a)$$

Second: $(-3 \to -3$ and 1, sum $-2)$

$$2ax^2 - 4a^2x - 6a^3 = 2a(x^2 - 2ax - 3a^2)$$
$$= 2a[x^2 - 3ax + ax - 3a^2]$$
$$= 2a[x(x - 3a) + a(x - 3a)]$$
$$= 2\boxed{a(x + a)}(x - 3a)$$

Third:

$$3(ax + a^2)^2 = 3[a(x + a)]^2$$
$$= 3a^2(x + a)^2 = 3\boxed{a(x + a)}a(x + a)$$

$$\text{H.C.F} = \boxed{a(x + a)} \text{ Ans}$$